

Let M be a set of poles which we take residues on (ex: $M = \{X_{16}, X_{27}, X_3\}$)

Let Z be a zero (defined by one or more linear eqs)

Def: Z is clean or strict w.r.t M if after imposing $X=0$ for all $X \in M$ the eqs defining Z remain nontrivial and do not force any additional active pole $Y \notin M$ to vanish.

equivalently, $\text{Res}_M(B|_Z) = (\text{Res}_M B)|_{Z, M}$.

ex:
$$\begin{cases} X_{24} = X_{14} - X_{13} \\ X_{25} = X_{15} - X_{13} \\ X_{26} = -X_{13} \end{cases}$$

$$B = \frac{a_1}{X_{24}} + \frac{a_2}{X_{25}} + \frac{a_3}{X_{26}} + \frac{b_1}{X_{13}} + \frac{b_2}{X_{14}} + \frac{b_3}{X_{15}}$$

$$B|_Z = \frac{a_1}{X_{14}-X_{13}} + \frac{a_2}{X_{15}-X_{13}} - \frac{a_3}{X_{13}} + \frac{b_1}{X_{13}} + \frac{b_2}{X_{14}} + \frac{b_3}{X_{15}}$$

$M = \{X_{13}\}$

$\text{Res}_{X_{13}} B|_Z = b_1 - a_3$

not clean

$(\text{Res}_{X_{13}} B)|_Z = b_1$

equivalently set $X_{13}=0$
the eqs say that $X_{26}=0$

$\text{Res}_{X_{14}} B|_Z = b_2$

$(\text{Res}_{X_{14}} B)|_Z = b_2$

clean

set $X_{14}=0$

$$\begin{aligned} X_{24} &= -X_{13} \\ X_{25} &= X_{15} - X_{13} \\ X_{26} &= -X_{13} \end{aligned}$$

Def: a zero is active for a given coefficient if it is clean after removing all terms that have already been shown to vanish.

more examples

z BP

$M = \{X_{14}, X_{15}\}$

set them to zero:

$Z \Rightarrow \begin{cases} X_{24} = -X_{13} \\ X_{25} = -X_{13} \\ X_{26} = -X_{13} \end{cases}$

$\Rightarrow$ clean (nothing is forced to vanish)

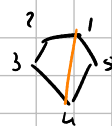
$M = \{X_{14}, X_{13}\}$

$Z \Rightarrow \begin{cases} X_{24} = 0 \\ \dots \end{cases} \Rightarrow$ not clean

5 point example

Residue at X_{14}

crossings: X_{25}, X_{35}



$$B_5 = \frac{a_1}{X_{13}X_{14}} + \frac{a_2}{X_{13}X_{24}} + \frac{a_3}{X_{13}X_{25}} + \frac{a_4}{X_{14}X_{24}} + \frac{a_5}{X_{14}X_{25}} + \frac{a_6}{X_{24}X_{25}} + \frac{a_7}{X_{24}X_{35}} + \frac{a_8}{X_{25}X_{35}} + \frac{a_9}{X_{14}X_{35}} + \frac{a_{10}}{X_{13}X_{35}}$$

residue at X_{14} : (same as multiply by X_{14} and set $X_{14} \rightarrow 0$)

$$= \frac{a_1}{X_{13}} + \frac{a_4}{X_{24}} + \frac{a_9}{X_{35}} + \frac{a_5}{X_{25}}$$

$\rightarrow B_4(1,2,3,4) \rightarrow$ the idea is that by induction we already know what happens to a_1, a_4 because we have also B_4 already

imply asserts: $B_4(1,2,3,4) = 0$

$$\frac{a_9}{X_{35}} - \frac{a_5}{X_{13}} = 0 \Rightarrow a_9 = a_5 = 0$$

$h=6$

Res at X_{15}

CROSSINGS X_{26}, X_{36}, X_{46}

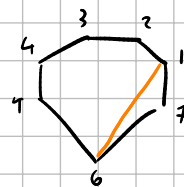
$$\begin{aligned} \rightarrow B_5(1,2,3,4,5) &+ \frac{1}{X_{26}} \left(\frac{a_{131526}}{X_{13}} + \frac{a_{141526}}{X_{14}} + \frac{a_{241526}}{X_{24}} + \frac{a_{251526}}{X_{25}} + \frac{a_{351526}}{X_{35}} \right) \\ &+ \frac{1}{X_{36}} \left(\frac{a_{131536}}{X_{13}} + \frac{a_{141536}}{X_{14}} + \frac{a_{241536}}{X_{24}} + \frac{a_{251536}}{X_{25}} + \frac{a_{351536}}{X_{35}} \right) \\ &+ \frac{1}{X_{46}} \left(\frac{a_{131546}}{X_{13}} + \frac{a_{141546}}{X_{14}} + \frac{a_{241546}}{X_{24}} + \frac{a_{251546}}{X_{25}} + \frac{a_{351546}}{X_{35}} \right) \\ &+ \frac{a_{263615}}{X_{26}X_{36}} + \frac{a_{264615}}{X_{26}X_{46}} + \frac{a_{364615}}{X_{36}X_{46}} \end{aligned}$$

7 point strategy

Residue on X_{16}

chords crossing X_{16} are $X_{27}, X_{37}, X_{47}, X_{57}$

4 chords per term



$$\begin{aligned} \text{Res}_{X_{16}=0} B_7 &= B_6^{(3)}(1,2,3,--,6) + \sum_{i=2}^5 \frac{C_{i7}^{(2)}(1,2,3,4,5,6)}{X_{i7}} + \sum_{2 \leq i < j < k \leq 5} \frac{C_{ijk7}^{(1)}(1,--,6)}{X_{i7}X_{j7}} \\ &+ \sum_{2 \leq i < j < k < l \leq 5} \frac{C_{ijkl7}^{(0)}(1,--,6)}{X_{i7}X_{j7}X_{k7}} \end{aligned}$$

goal: $C_{i7}^{(2)} = 0 \quad C_{ijk7}^{(1)} = 0 \quad C_{ijkl7}^{(0)} = 0$

which 1-zero do we use?

X_{16} must not be in the eq. with only 2 terms (otherwise not a clean zero)

$$X_{24} = X_{14} - X_{13}$$

$$X_{25} = X_{15} - X_{13}$$

$$X_{26} = X_{16} - X_{13}$$

$$X_{27} = -X_{13}$$

born free: $X_{35} \ X_{36} \ X_{32}$

$X_{46} \ X_{47}$

X_{57}

suppose we use this instead

$$X_{13} = X_{73} - X_{72}$$

$$X_{14} = X_{74} - X_{72}$$

$$X_{15} = X_{75} - X_{72}$$

$$X_{16} = -X_{72}$$

$$\rightarrow C_{345}^{(n)} = 0$$

other reason why good choice of 1-zero: if we remove everything that has a 7 and set X_{16} to zero, the constraints gives us

$$X_{24} = X_{14} - X_{13}$$

$$X_{25} = X_{15} - X_{13}$$

$$X_{26} = -X_{13}$$

$X_{35} \ X_{36} \ X_{46}$
↓ free

→ same 1-zero for $n=6$.

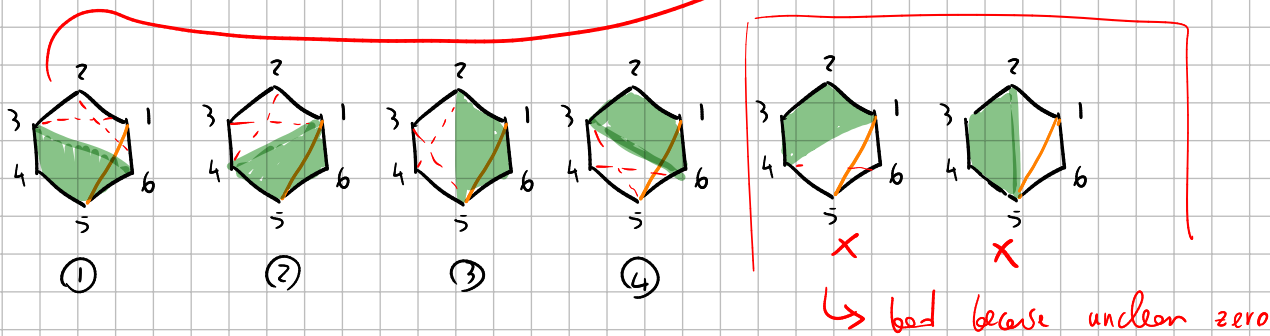
Prop: if $B_n = 0 \ \forall$ hidden zeros, then B_n is balanced and unkey. $\equiv P_n$

induction hypothesis: P_{n-1} true: iff $B_{n-1} = 0 \ \forall$ hidden zero then B_{n-1} is balanced and unkey.

show that P_n is true.

How? suppose $B_n = 0$

$$\begin{aligned}
& B_5(1,1,1,4,5) + \frac{1}{x_{26}} \left(\frac{a_{13\ 15\ 26}}{x_{13}} + \frac{a_{14\ 15\ 26}}{x_{14}} + \frac{a_{24\ 15\ 26}}{x_{24}} + \frac{a_{25\ 15\ 26}}{x_{25}} + \frac{a_{35\ 15\ 26}}{x_{35}} \right) \\
& + \frac{1}{x_{36}} \left(\frac{a_{13\ 15\ 36}}{x_{13}} + \frac{a_{14\ 15\ 36}}{x_{14}} + \frac{a_{24\ 15\ 36}}{x_{24}} + \frac{a_{25\ 15\ 36}}{x_{25}} + \frac{a_{35\ 15\ 36}}{x_{35}} \right) \\
& + \frac{1}{x_{46}} \left(\frac{\cancel{a_{13\ 15\ 46}}}{\cancel{x_{13}}} + \frac{\cancel{a_{14\ 15\ 46}}}{\cancel{x_{14}}} + \frac{\cancel{a_{24\ 15\ 46}}}{\cancel{x_{24}}} + \frac{\cancel{a_{25\ 15\ 46}}}{\cancel{x_{25}}} + \frac{\cancel{a_{35\ 15\ 46}}}{\cancel{x_{35}}} \right) \\
& + \frac{\cancel{a_{26\ 36\ 15}}}{\cancel{x_{26}x_{36}}} + \frac{a_{26\ 46\ 15}}{x_{26}x_{46}} + \frac{a_{36\ 46\ 15}}{x_{36}x_{46}}
\end{aligned}$$



$$a_{36\ 46\ 15} = 0 \quad \text{by } \textcircled{1}$$